

KAPIL RAVI RATHOD

☎ +1 (582) 203-8438 ✉ kapilrathod1234@gmail.com 📁 [Portfolio](#) [LinkedIn](#) [GitHub](#)

EDUCATION

The Pennsylvania State University, University Park

Aug 2024 – May 2026 (Expected)

GPA: 3.9 | *M.Sc. in Computer Science and Engineering*

State College, PA

Assistantships: TA for CMPSC 132 - Programming and Computation II: Data Structures

National Institute of Technology, Warangal

Aug 2017 – May 2021

CGPA: 7.98 | *B.Tech. in Computer Science and Engineering*

Warangal, TS

TECHNICAL SKILLS

Programming Languages: Python, Java, C/C++, Go, R, JavaScript, Bash, CUDA C++

Databases: MySQL, PostgreSQL, Oracle, MongoDB, Redis, Qdrant, FAISS, Milvus

Dev Tools: Spring Boot, Angular.js, Node.js, React.js, Next.js, Django, Docker, Git, Hadoop, Kafka, Spark, gRPC

Testing & Cloud: PyTest, Unittest, Locust, Behave, Selenium, JUnit, GCP [Link], Prefect, OpenShift, Jenkins, Kubernetes

AI/ML Tools: PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, LangChain, AutoGen, LangGraph, DSPy, DeepEval

Visualization Tools: Looker, Hex

EXPERIENCE

Recursion

May 2025 – Aug 2025

SWE Intern

Salt Lake City, UT

- Delivered **4 new Django and React/Next.js components features** for the Experiment Configuration orchestration engine.
- Streamlined Prefect pipelines, automating data ingestion to GCP cloud storage, **improving operational efficiency by 60%**.
- Supported **3 critical** dashboards (Hex, Looker) through diverse visualizations, enabling data-driven insights from RADME data.

Deutsche Bank

Jul 2021 – Jul 2024

SDE2 - Associate Software Engineer

Pune, MH

- Architected a RAG-powered multi-LLM chatbot unifying internal documentation; used by **900+ users** and saving **\$3M/year**.
- Engineered an LLM-based alerting & monitoring service for global regulatory tracking, **saving \$1M/year and 700+ hours**.
- Implemented a Plotly dashboard for **100+ users** to monitor and manage DaS platform computational resource usage per user.

SDE1 - Graduate Analyst

- Automated UC4 pre-deployment checks with Python/UNIX to parse config files, **reducing manual effort by 90%**.
- Enhanced a bank-wide translation service for intranet webpages, **boosting translation quality (BLEU scores)** by **12%**.
- Improved a reporting app's infra, security and UI; created dashboards that **saved 30+ hours/week** on manual updates.
- Built a BERT-based automated resume matching tool, **saving 80% of manual effort across 100+ hiring managers**.

Semut.io

Apr 2020 – Jul 2020

Cloud Native Engineering Intern

Remote

- Contributed to a Go-based application manager by implementing **3 core features** for Elasticsearch-as-a-service provisioning.

HERE Technologies

May 2019 – Jul 2019

SWE Intern

Mumbai, MH

- Designed a UI-based RESTful bulk-upload POI spell-checking service **saving 100+ hours** for the map creation team.
- Ideated a POC for stitching Here vehicle Lidar images, enhancing autonomous vehicle perception **by processing 1,000+ frames**.

PROJECTS

Fault-Tolerant Distributed Key-Value Store with Paxos | *Go, RPC, Paxos*

Jan 2025 – Mar 2025

- Designed a log-replicated distributed key-value store using Paxos, ensuring **100% consistency** across replicas and reducing duplicate client execution by **95%** via idempotence checks.
- Achieved **99.9% uptime** and fault-tolerance by recovering and synchronizing lagging replicas using consensus replays.

Hybrid GA + RL XSS Fuzzer for DOM-Based Attacks | *Python, Selenium*

Jan 2025 – May 2025

- Proposed a hybrid GA+RL XSS fuzzer with a custom risk-aware fitness function, discovering **50+ novel stealthy XSS payloads** on standard testbeds to expose DOM-based vulnerabilities.

Systolic Array Optimization for Neural Networks | *ScaleSim V2*

Oct 2024 – Dec 2024

- Tuned systolic arrays with ScaleSim V2 to optimize performance, utilization and area-delay for neural networks.
- Identified 64×64 array with Output Stationary dataflow as optimal across AlexNet, YOLO Tiny and MobileNet variants.

CPU Architecture Sensitivity Analysis using GEM5 | *C++, GEM5*

Sep 2024 – Oct 2024

- Simulated **30+ GEM5 CPU-memory configurations** and benchmarked Stanford Quicksort (4.45T total ticks).
- Analyzed cache miss, access latency and power across configurations to inform CPU architecture sensitivity.

PUBLICATIONS

K.R. Rathod, S. Bansal, Y.H.C. Kandpal and M. Bisi *Learning Automata-based Feature Selection for Duplex Output Software Effort Estimation Model*, 12th ICCNT, Kharagpur, India, 2021. [Link]